

Atom-Resolved Structural Ring Boundary Specification

Stage 11C3: Persistent T/O Chemistry, Serration, Harmonics, and Dihedral Gauges

mdstats

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Purpose and stage boundary

Stage 11C3 extends the persistent T/O polygons from Stages 11C1 and 11C2 into an atom-resolved, species-independent structural boundary:

```
reference T/O ring identities + compatible-frame ring geometry
+ immutable framework chemistry
+ optional exact-source crystallographic aliases
-> ordered T/O atom records
-> exact cyclic-index structural sequences
-> unweighted cyclic spectra
-> boundary-measure angular moments
-> rank-safe physical-angle fits
-> symmetry-breaking, class-splitting, and continuity diagnostics
```

Runtime/API target:

mdstats 0.19.94a0

Primary module:

mdstats/analysis/ring_boundary.py

The exact ordered Cartesian T/O coordinates remain authoritative. Harmonic coefficients are lossy descriptors and never replace, smooth, reconstruct, or reorder the source ring.

This stage owns framework geometry and chemistry only. It does **not** own:

- M–O or M–T distances;
- mobile-species coordination fingerprints;

- statistical site centers or labels;
- occupancy-conditioned structural states;
- registered structural views; or
- kinetic transitions.

Those responsibilities remain downstream in C0A3 and Stage 11E5a.

External methods and original construction

The finite cyclic spectrum is the standard discrete Fourier transform (DFT), using the conventional negative exponential [1]. Weighted physical-angle regression uses standard weighted linear least squares and condition/rank analysis [2]. Continuous phase uncertainty uses first-order covariance propagation; later blockwise phase distributions should use circular statistics rather than ordinary linear means [3].

The following are project-specific `mdstats` constructions:

- strict separation of equal-atom cyclic DFTs from boundary-measure moments;
- exact even-ring Nyquist handling as a signed real coordinate;
- persistent ring-atom and periodic-image chemistry records;
- source-bound, fail-closed LTA $O(1)/O(2)/O(3)$ alias profiles;
- explicit radial-center provenance and singular-angle rejection;
- simultaneous equal-atom and arc-length-weighted actual-angle fits;
- dihedral gauge metadata and coefficient transformation utilities;
- per-frame oxygen-class splitting, symmetry-breaking, and reference continuity;
- deterministic source replay, strict JSON serialization, and resource preflight.

Inputs and source compatibility

The public constructor is:

```
build_structural_ring_boundary_catalog(
    reference_geometry: ReferenceRingGeometryCatalog,
    frame_geometry: FrameRingGeometryCatalog,
    collection: AtomisticFrameCollection,
    *,
    alias_profile: LtaOxygenAliasProfile | None = None,
    options: RingBoundaryOptions | None = None,
    resources: RingBoundaryResources | None = None,
) -> StructuralRingBoundaryCatalog
```

The frame catalog must be bound to the supplied reference-ring digest. The atomic-number vector is hashed separately as `collection_chemistry_digest`. This stage does not infer or repair chemistry from coordinates.

The implementation preflights:

- number of persistent rings;
- number of compatible frames; and
- predicted T/O atom-record count.

A resource violation raises before descriptor arrays are allocated.

Persistent atom boundary

For a ring of order k , Stage 11C3 retains two aligned sequences:

$$(T_0, T_1, \dots, T_{k-1}), \quad (O_0, O_1, \dots, O_{k-1}),$$

where O_j bridges T_j and $T_{j+1 \bmod k}$. Every atom record contains:

```
boundary kind: T or O
cyclic index and canonical orientation
persistent atom index and lattice-image shift
atomic number and element symbol
```

reference Cartesian coordinate
 instantaneous Cartesian coordinate
 local (u, v, n) coordinate
 projected radius, polar angle, and normal coordinate

An oxygen record additionally contains:

ordered neighboring T references
 ordered neighboring T atomic numbers and Si/Al classes
 generic orientation-independent oxygen environment signature
 optional validated crystallographic alias

The generic environment signature is based on the two neighboring T classes. It remains available when no framework-specific alias is admissible.

Center and local-coordinate contract

The default center is the Stage-11C1 projected oxygen-area centroid:

$$\mathbf{c}_R = \mathbf{c}_{O,A}.$$

The local frame is the first persistent side frame $(\hat{\mathbf{e}}_u, \hat{\mathbf{e}}_v, \hat{\mathbf{n}})$. For atom coordinate $\mathbf{x}_j$,

$$u_j = (\mathbf{x}_j - \mathbf{c}_R) \cdot \hat{\mathbf{e}}_u, \quad v_j = (\mathbf{x}_j - \mathbf{c}_R) \cdot \hat{\mathbf{e}}_v, \quad z_j = (\mathbf{x}_j - \mathbf{c}_R) \cdot \hat{\mathbf{n}},$$

$$\rho_j = \sqrt{u_j^2 + v_j^2}, \quad \theta_j = \text{atan2}(v_j, u_j).$$

Every ring record stores:

center_kind = oxygen_area_centroid
 center_coordinates
 center_uncertainty
 minimum_projected_radius
 angular_coordinate_defined

The current deterministic geometry has center_uncertainty = 0. This is not a statistical uncertainty claim; it means that no stochastic center estimator is used at this stage.

If any required projected radius satisfies

$$\rho_j \leq \rho_{\min},$$

actual-angle fits fail closed with angular-coordinate-undefined. No arbitrary angle is assigned to the singular atom, and no boundary-measure moment is emitted for that affected sequence.

Exact ordered structural sequences

Every resolved boundary provides exact sequences for:

- T projected radius;
- T normal coordinate;
- T atomic number;
- O projected radius;
- O normal coordinate; and
- number of adjacent Al T atoms for each O.

These arrays preserve cyclic order. String-valued environment signatures and optional aliases remain in atom records and are not silently converted into an arbitrary numerical code.

Unweighted cyclic-index spectrum

For a real ordered scalar sequence $y_j, j = 0, \dots, k-1$,

$$\tilde{y}_m = \frac{1}{k} \sum_{j=0}^{k-1} y_j \exp\left(-\frac{2\pi i m j}{k}\right).$$

UnweightedCyclicIndexSpectrum always uses equal atom weights. It stores only the unique real-sequence modes

$$m = 0, 1, \dots, \left\lfloor \frac{k}{2} \right\rfloor.$$

For even k , $m = k/2$ is the Nyquist mode. Its coefficient is real up to floating-point cleanup and is represented by:

```
amplitude
nyquist_orientation_sign in {-1, 0, +1}
phase_defined = false
```

It does not own a continuous phase. In particular, the S6R sequence

$$a, b, a, b, a, b$$

has only a mean and an exact signed $m = 3$ cyclic component.

Each non-Nyquist mode stores:

```
real and imaginary coefficient
raw amplitude
optional normalized amplitude
phase when amplitude support is sufficient
phase_defined
phase_uncertainty
```

For an exact deterministic DFT descriptor, a resolved phase has numerical uncertainty zero. Temporal/statistical phase uncertainty belongs to later aggregation.

Dihedral gauge

The canonical catalog uses the inherited ring origin and orientation. Public utilities expose the exact dihedral action for testing and downstream matching.

If a new cyclic origin begins at old position s without reversal,

$$y'_j = y_{j+s}, \quad \tilde{y}'_m = \tilde{y}_m e^{+2\pi i m s/k}.$$

If orientation is reversed about old position s ,

$$y'_j = y_{s-j}, \quad \tilde{y}'_m = \tilde{y}_m^* e^{-2\pi i m s/k}.$$

The module provides:

```
apply_cyclic_dihedral_gauge(...)
transform_cyclic_coefficient(...)
```

The ring record retains the canonical origin atom, orientation, and reversal origin. A phase-dependent classification must reject an unresolved phase.

Boundary-measure angular moments

A physical corrugated boundary is sampled nonuniformly in angle and arc length. For positive boundary weights w_j ,

$$M_m^{(\text{bdry})} = \frac{\sum_j w_j y_j e^{-im\theta_j}}{\sum_j w_j}.$$

The implemented default is the polygonal arc-length Voronoi weight

$$w_j = \frac{1}{2} (\|\mathbf{x}_j - \mathbf{x}_{j-1}\| + \|\mathbf{x}_{j+1} - \mathbf{x}_j\|).$$

This output is named `BoundaryMeasureAngularMoments`. It is not called an unweighted DFT and makes no claim of exact cyclic orthogonality or exact Nyquist semantics.

Rank-safe physical-angle fits

For actual polar angles θ_j and requested modes $\mathcal{M}$,

$$y_j = c_0 + \sum_{m \in \mathcal{M}} [a_m \cos(m\theta_j) + b_m \sin(m\theta_j)] + \epsilon_j.$$

The design has

$$p = 1 + 2|\mathcal{M}|$$

real parameters. Before solving, the implementation records and validates:

- angular-coordinate admissibility;
- sample count and design rank;
- weighted-design condition number;
- requested weighting measure;
- declared regularization; and
- requested mode set.

A fit is unresolved when:

```
rank < parameter_count
condition_number > maximum_condition_number
an angular coordinate is undefined
```

No pseudoinverse is used to manufacture non-identifiable coefficients. Regularization does not override the full-rank requirement.

For every geometric sequence, the catalog attempts two separately named fits:

```
equal_atom
arc_length_voronoi
```

They are retained as distinct estimators.

For $g_m = a_m - ib_m$, the physical-angle phase is

$$\phi_m = \arg g_m.$$

When residual degrees of freedom exist, first-order covariance propagation provides `phase_uncertainty`. With no residual degrees of freedom, the phase may be resolved but its uncertainty remains unavailable rather than being reported as zero.

Raw and normalized amplitudes

Raw amplitudes retain the units of the source sequence. A normalized amplitude is emitted only when a positive declared scale exists. For radial signals the default scale is the absolute mean radius,

$$\bar{A}_m = \frac{A_m}{|\bar{\rho}|},$$

provided $|\bar{\rho}|$ exceeds `normalization_floor`.

A missing or inadmissible scale produces

```
normalization_admissible = false
normalized_amplitude = null
```

A nonpositive explicitly supplied normalization scale is an input error.

Oxygen-class splitting and symmetry breaking

O atoms are grouped by validated crystallographic alias when available; otherwise they are grouped by the generic neighboring-T environment signature. For every class, the result stores count, mean radial coordinate, and mean normal coordinate.

The maximum class radial split is

$$\Delta\rho_{\max} = \max_c \bar{\rho}_c - \min_c \bar{\rho}_c.$$

The oxygen radial symmetry-breaking diagnostic is

$$S_\rho = \frac{\sqrt{\sum_{m>0} |\tilde{\rho}_m|^2}}{|\tilde{\rho}_0|},$$

when the mean radius is admissible. The dominant radial mode is reported only when its amplitude exceeds the declared support tolerance.

These are descriptors, not site classifications.

Reference-to-frame continuity

Every compatible-frame boundary retains the same persistent atom identities. For each cyclic sequence and mode, the frame result compares with the reference:

- amplitude ratio when the reference amplitude is nonzero;
- phase difference modulo $2\pi/m$ for resolved non-Nyquist phases; or
- Nyquist sign support for even-ring Nyquist modes.

No phase difference is reported when either amplitude is unresolved.

LTA crystallographic aliases

LtaOxygenAliasProfile is optional and exact-source-bound. It contains:

```
profile_id
reference_ring_geometry_digest
atom-index -> 0(1)/0(2)/0(3) mapping
complete-coverage policy
optional S6R 0(2)/O(3) alternation validation
```

Validation fails closed if:

- the reference digest differs;
- a mapped atom is not a persistent ring oxygen;
- an alias is assigned to a non-oxygen atom;
- required coverage is incomplete; or
- requested S6R O(2)/O(3) alternation is violated.

Exact digest binding is intentional. A conventional crystallographic alias must not be transferred silently across a different atom ordering, origin, framework composition, or structural model.

Persistent outputs and serialization

StructuralRingBoundaryCatalog retains:

```
reference and compatible-frame geometry digests
collection chemistry digest
alias validation and profile provenance
options and resource contract
reference boundaries
framewise boundaries
canonical schema and digest algorithm
canonical SHA-256 digest
```

`from_dict(...)` rebuilds from the supplied scientific sources and requires byte-equivalent canonical JSON content. Tampered derived values are rejected.

Acceptance tests

The focused C3 gate requires:

1. exact S6R alternating-sequence recovery at cyclic $m = 3$;
2. declared cyclic shift and reversal coefficient transforms;
3. irregular-angle alternation not being forced into a pure physical-angle $m = 3$ fit;
4. rank-deficient actual-angle mode sets failing closed;
5. singular projected atoms making physical-angle fits unresolved;
6. boundary-measure moments remaining distinct from unweighted DFTs;
7. phase becoming undefined below amplitude support;
8. all 58 real LTA ring boundaries retaining persistent T/O atom identities;
9. rigid-rotation invariance of local sequences and harmonic amplitudes;
10. optional aliases attaching only after exact-source validation;
11. source-mismatched and incomplete aliases failing closed;
12. serialization replay, resource preflight, and public exports.

Deferred boundaries

The following remain outside Stage 11C3:

- C0A3 application of affine registration to structural views;
- site-conditioned M–O/M–T fingerprints;
- geometry-forward prediction of mobile-ion coordination;
- off-center versus serration residual classification;
- occupancy-conditioned exchangeability;
- circular block statistics over long trajectories; and
- transition-path sector analysis.

References

- [1] A. V. Oppenheim and R. W. Schaffer, *Discrete-Time Signal Processing*, 3rd ed., Pearson, 2010.
- [2] G. H. Golub and C. F. Van Loan, *Matrix Computations*, 4th ed., Johns Hopkins University Press, 2013.
- [3] K. V. Mardia and P. E. Jupp, *Directional Statistics*, Wiley, 2000.